

FINAL PROJECT DOCUMENT

Smartphone Addiction Analytics Dashboard

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Team ID	MCA
Project Name	Smartphone Addiction Analytics Dashboard
Document Type	Final Project Submission – All Phases Consolidated

This document consolidates all project phases: Problem Definition, Project Planning, Proposed Solution, Data Collection & Preprocessing, Data Quality, Business Questions & Visualizations, Dashboard Design, and Report Design.

1. Project Overview

1.1 Objective

Analyze demographic, behavioural and lifestyle attributes from smartphone usage data to uncover patterns, identify high-risk user segments, and determine the strongest predictors of smartphone addiction.

1.2 Scope

Build an interactive analytical dashboard covering age, gender, daily/weekend screen time, social media & gaming hours, notifications, app opens, sleep, stress, academic/work impact and addiction level. Focus is on descriptive and diagnostic analytics for product, research and well-being stakeholders. Predictive modelling and real-time mobile integration are out of scope.

1.3 Problem Statement

Smartphone usage behaviour is driven by many interacting factors (screen time, notifications, sleep, stress, academic impact). Without a consolidated view, product teams, researchers and digital well-being programmes overlook high-risk segments, misjudge which behaviours drive addiction, and miss opportunities for early intervention.

Impact: Solving this enables stakeholders to reduce the share of users in the Severe addiction band, prioritise feature investment (usage limits, mindful notifications), design targeted well-being nudges, flag early warning signs, and provide evidence-based digital well-being thresholds.

2. Define Problem Statements (Customer Problem Statement Template)

A well-articulated customer problem statement allows the team to find the ideal solution for customers' challenges and empathize with how they perceive the product or service.

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Maya Chen, Head of Digital Well-Being Product	understand how screen time, social media hours & notifications relate to addiction level across users	I cannot clearly identify which user segments carry the highest risk of unhealthy usage	there is no consolidated interactive view combining demographic, behavioural and lifestyle data	unable to prioritise feature investment, design targeted nudges, or report effectively to leadership
PS-2	Daniel, User Behaviour Analyst	analyse how gaming, work/study, sleep hours and stress influence the likelihood of addicted usage	early behavioural warning signs remain hard to detect across the population	lifestyle risk factors are scattered and not easily flaggable in one place	limited ability to support intervention programmes such as sleep-hygiene reminders and stress-aware coaching

3. Project Proposal (Proposed Solution)

3.1 Approach

Load and clean the Smartphone Usage dataset (7,500 records). Perform exploratory data analysis and correlation analysis. Build interactive visualisation and filters (age, gender, stress, academic impact). Deliver persona-specific views for product managers, behaviour analysts and digital-health researchers.

3.2 Key Features

1. Overview KPIs (Severe addiction %, average screen time, high-stress users).
2. Segment comparison by age / gender / stress / academic impact.
3. Behaviour insight charts (gaming, sleep, work/study vs addiction).
4. Predictor ranking (app opens, notifications, weekend screen time).
5. Interactive filters and drill-down for high-risk segment identification.

3.3 Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	CPU for data processing and dashboard serving	4–8 core CPU (no GPU required)
Memory	RAM for data loading and interactive queries	16 GB RAM
Storage	Disk space for dataset, processed files and logs	50 GB SSD
Frameworks	Dashboard & analytics frameworks	Streamlit / Dash / Power BI or Tableau
Libraries	Data analysis & visualisation libraries	pandas, numpy, plotly, seaborn, scikit-learn
Dev Environment	IDE and version control	VS Code / Jupyter Lab, Git + GitHub
Data	Source, size and format of primary dataset	Smartphone_Usage.csv – 7,500 records, 16 columns (CSV)

4. Product Backlog, Sprint Schedule & Estimation

The project is delivered across three sprints with a total of 26 story points.

Sprint	Epic	USN	User Story / Task	Points	Priority	Team	Dates
Sprint-1	Data Foundation	USN-1	As Maya/Daniel/Priya, I can load and clean the Smartphone_Usage dataset so that all	3	High	Data Engineer	22–28 Sep 2026

			key metrics are available for analysis				
Sprint-1	Dashboard Overview	USN-2	As Maya Chen, I can view high-level KPIs (Severe addiction %, avg screen time, high-stress users) on an overview page	5	High	Frontend Dev	22–28 Sep 2026
Sprint-2	Segment Analysis	USN-3	As Maya Chen, I can filter and compare addiction levels by age group, gender, stress level and academic impact	5	High	Frontend + Analyst	29 Sep – 05 Oct
Sprint-2	Behavioural Insights	USN-4	As Daniel, I can explore relationships between gaming, sleep, work/study hours and addiction likelihood with interactive charts	5	High	Frontend + Analyst	29 Sep – 05 Oct
Sprint-3	Predictors & Recommendations	USN-5	As Dr. Priya Nair, I can view the strongest predictors of addiction and receive evidence-based well-being thresholds	8	Medium	Full Team	06–12 Oct 2026

5. Data Collection Plan & Raw Data Sources

5.1 Project Overview (Data)

The project aims to build an interactive analytical dashboard that combines demographic, behavioural and lifestyle attributes from smartphone usage data to uncover patterns of digital addiction, identify high-risk user segments, and support product, research and well-being stakeholders.

5.2 Data Collection Plan

Primary data will be collected from the provided Smartphone_Usage.csv file. No additional external APIs or real-time streams are required for the current scope.

Future enhancements may include anonymised mobile telemetry or survey responses, but these are out of scope for this phase.

5.3 Raw Data Sources Identified

Source Name	Description	Location	Format	Size	Access
Smartphone_Usage.csv	Structured dataset of 7,500 young-adult users (age 18–35) containing demographics, daily/weekend screen time, social media & gaming hours, notifications, app opens, sleep, stress, academic impact and addiction labels	Local project repository /attachments/	CSV	~0.6 MB (7,500 × 16)	Team internal – full read

No secondary external dataset is required for the current project scope.

6. Data Exploration and Preprocessing

6.1 Data Overview

Smartphone_Usage.csv contains 7,500 records and 16 columns describing young adult users (age 18–35). Key fields: demographics (age, gender), usage behaviour (daily/weekend screen time, social media, gaming, work/study hours, notifications, app opens), lifestyle (sleep hours, stress level, academic/work impact) and target variables (addiction_level, addicted_label). Source is a structured CSV suitable for descriptive and diagnostic analytics.

6.2 Data Cleaning

No duplicate rows found (0 duplicates). Only addiction_level has missing values (819 NaNs ≈ 10.9%). Plan: (1) keep the rows for completeness of other metrics, (2) create a binary flag is_addiction_missing, (3) for modelling/dashboard filters treat missing as a separate category 'Unknown'. All other columns are complete. No obvious data-entry errors detected in numeric ranges.

6.3 Data Transformation

In Power Query / pandas: (1) filter out any future invalid ranges if needed, (2) create calculated columns – `total_entertainment_hours = social_media_hours + gaming_hours`, `screen_to_sleep_ratio = daily_screen_time_hours / sleep_hours`, `high_notification_flag = notifications_per_day > 150`. (3) Pivot / group-by summaries for age-group and gender will be created for dashboard visuals. Sorting by addiction severity for priority views.

6.4 Data Type Conversion

age, notifications_per_day, app_opens_per_day, addicted_label already integer. All hour columns already float. Categorical columns (gender, stress_level, academic_work_impact, addiction_level) converted to category / ordered category where appropriate (e.g. addiction_level ordered as None < Mild < Moderate < Severe). transaction_id and user_id kept as string identifiers.

6.5 Column Splitting and Merging

No splitting required – all columns are atomic. Optional merge: create a derived dimension table for user demographics (user_id, age, gender) if moving to a star-schema model later. For current single-table analysis no merge is necessary.

6.6 Data Modeling

Single fact table is sufficient for the dashboard. If expanded later: User dimension (user_id, age, gender) linked to Fact_Usage on user_id. Measures planned: % Severe addiction, Avg daily screen time, Avg notifications, High-risk user count, Correlation coefficients between key numeric predictors and addicted_label.

6.7 Save Processed Data

Cleaned dataset saved as Smartphone_Usage_Cleaned.csv (and optionally .parquet). A second version with imputed / flagged addiction_level and calculated columns is stored as Smartphone_Usage_Processed.csv for dashboard consumption. Version control via Git; original raw file retained unchanged.

7. Data Quality Report

The Data Quality Report summarizes data quality issues from the selected source, including severity levels and resolution plans.

Data Source	Data Quality Issue	Severity	Resolution Plan
Smartphone_Usage.csv	Missing values in addiction_level column (819 records ≈ 10.9%)	Moderate	Retain rows; create flag is_addiction_missing; treat missing as

			category 'Unknown' in filters and visuals. Do not impute for dashboard accuracy.
Smartphone_Usage.csv	No duplicate records found (0 duplicates across 7,500 rows)	Low	No action required. Confirm uniqueness of transaction_id and user_id remains intact after any future joins.
Smartphone_Usage.csv	Categorical consistency: gender (Male/Female/Other), stress_level (Low/Medium/High), academic_work_impact (Yes/No) – all values valid	Low	Convert to ordered categorical types. Enforce domain validation in Power Query / pandas to prevent future invalid entries.
Smartphone_Usage.csv	Numeric ranges appear clean (screen time 3–12 hrs, sleep 4.5–9 hrs, notifications 20–250). No extreme outliers detected beyond expected behaviour	Low	Apply soft upper/lower bounds only if business rules require (e.g. screen time > 16 hrs). Currently no clipping needed.
Smartphone_Usage.csv	addicted_label is binary (0/1) while addiction_level is ordinal (None/Mild/Moderate/Severe) – slight conceptual overlap	Low	Keep both columns. Use addiction_level for detailed analysis and addicted_label for binary classification views. Document the relationship clearly.

8. Business Questions and Visualization Report

The following business questions are framed specifically around the available columns in the Smartphone_Usage dataset. Each question is paired with a recommended Power BI visualization.

1. What is the average daily screen time across different age groups?

Visualization: Column Chart comparing average daily_screen_time_hours by age brackets (18-22, 23-27, 28-32, 33-35).

2. How does addiction level vary by gender?

Visualization: Stacked Bar Chart or 100% Stacked Column showing distribution of addiction_level (None, Mild, Moderate, Severe) across Male, Female, and Other.

3. What is the relationship between daily screen time and sleep hours?

Visualization: Scatter Plot with daily_screen_time_hours on X-axis and sleep_hours on Y-axis, colored by addiction_level.

4. Which stress levels are most associated with high addiction?

Visualization: Clustered Bar Chart showing count of users by stress_level and addiction_level.

5. How does weekend screen time compare to weekday daily screen time?

Visualization: Clustered Column Chart comparing average daily_screen_time_hours vs average weekend_screen_time.

6. What is the impact of high social media hours on academic/work performance?

Visualization: Box Plot or Column Chart of social_media_hours grouped by academic_work_impact (Yes/No).

7. How do notification volume and app opens correlate with addiction label?

Visualization: Scatter Plot of notifications_per_day vs app_opens_per_day, colored by addicted_label (0 vs 1).

8. What is the distribution of addiction levels across the entire user base?

Visualization: Donut Chart or Pie Chart showing percentage of users in None, Mild, Moderate, and Severe addiction_level.

9. Dashboard Design

Creating an effective dashboard involves thoughtful design to ensure that the presented information is clear, relevant, and easily understandable for the intended audience.

9.1 Design Principles Applied

- Clear and Intuitive Layout – KPI cards at the top, followed by supporting charts in a clean grid
- Appropriate Visualizations – Bar charts, Pie charts, and horizontal bars chosen based on the nature of the metric
- Colour and Theming – Consistent blue palette for visual harmony and easy interpretation
- Interactive Filters – Addiction Level slicer allows users to filter the entire dashboard
- Drill-Down Capabilities – Charts support interaction for deeper exploration of segments
- Responsive Design – Layout adapts for different screen sizes

9.2 Major Outcomes from the Dashboard

1. **Total Users Analysed:** The dashboard covers 7,500 users (Count of user id), providing a solid sample size for insights.
2. **High Addiction Prevalence:** 5,308 users are classified as addicted (addicted_label = 1), indicating that approximately 70.8% of the user base shows signs of addiction.
3. **Weekend Screen Time:** Average weekend screen time is 9.24 hours, which is notably high and higher than typical weekday usage for many users.
4. **Average Sleep Hours:** Users average 6.74 hours of sleep, which is below the recommended 7–9 hours, suggesting a potential link between high screen time and reduced sleep.
5. **Gender & Screen Time:** Average daily screen time is very similar across genders (Male 7.6 hrs, Other 7.5 hrs, Female 7.4 hrs), showing no major gender disparity in overall usage.
6. **Addiction Level Distribution:** The pie chart shows Moderate ($\approx 32\%$), Severe ($\approx 18\%$), Mild ($\approx 18\%$), None ($\approx 11\%$) (with some missing), confirming that Moderate and Severe together form the majority of labelled users.
7. **Sleep by Addiction Level:** Severe addiction users have slightly higher average sleep hours than expected, while Mild users show the lowest – indicating complex relationships that need further investigation.
8. **Weekend Screen Time by Addiction Status:** Addicted users average 10.2 hours of weekend screen time versus 6.9 hours for non-addicted users – a clear behavioural differentiator.
9. **Age Group Distribution:** The largest age segment is 20–29 years ($\approx 49\%$), followed by 30–39 and Under 20, showing the dashboard is focused on young adults.
10. **App Opens by Stress Level:** Average app opens per day are very similar across stress levels (Medium 98.7, Low 97.5, High 97.3), suggesting that stress does not strongly differentiate app-opening behaviour in this dataset.

10. Report Design

A report is a comprehensive document that provides a detailed and structured account of data analysis, findings, and insights. Observations drawn from the reports provide valuable insights into digital addiction patterns and screen-time behaviour.

10.1 Report 1 – Digital Addiction & Screen Time Behavior Analysis

1. **High App Engagement:** Average app opens per day is 97.83, indicating very frequent phone checking behaviour across the user base.
2. **Social Media Usage:** Average social media hours is 3.27 hours per day – a significant portion of total screen time.
3. **High Notification Load:** Users receive an average of 134.26 notifications per day, which is a major driver of habitual checking.

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4. **Addiction Level Counts:** Moderate addiction has the highest count (2,874), followed by Severe (2,434), Mild (1,373) and None (819). Moderate + Severe together form the majority of labelled users.
 5. **Work/Study Hours vs Academic Impact:** Users with and without academic/work impact show almost identical average work/study hours (~3.2–3.3 hrs), suggesting that screen time outside study hours may be the real differentiator.
 6. **Weekend Screen Time by Addiction:** Severe users average 10.4 hrs and Moderate users 10.1 hrs of weekend screen time, while Mild (7.3) and None (6.2) are significantly lower – a clear behavioural separator.
 7. **Age Group Trend:** Average weekend screen time slightly declines from 20–29 to Under 20 age groups, with 20–29 being the highest usage segment.

10.2 Report 2 – Smartphone Addiction & Digital Well-Being Insights

1. **Gaming Hours:** Average gaming hours is 2.01 hours per day – a meaningful contributor to total entertainment time.
2. **User Demographics:** Average age of users is 26.57 years, confirming the dataset focuses on young adults (18–35).
3. **stress Level Distribution:** Stress levels are almost evenly split – High (~34%), Low (~34%), Medium (~32%), showing stress is widespread across the population.
4. **Sleep Hours Decline with Age:** Average sleep hours drop from ~6.76 (20–29) to lower values in the 30–39 group, indicating older young-adults sleep slightly less.
5. **High Screen Time Users:** Approximately 34% of users fall into the High Usage category, while 66% are in Normal Usage – a useful segmentation for intervention targeting.
6. **Social Media Hours by Addiction:** Severe users average 3.8 hrs and Moderate users 3.6 hrs of social media, while Mild (2.3) and None (2.2) are substantially lower – social media intensity is strongly linked to addiction severity.
7. **Key Influencers of Addiction (from AI visual):** Daily screen time increase of 2.61 hrs raises addiction likelihood by 25.96×; social media hours increase of 1.59 hrs raises it by 10.24×; weekend screen time decrease of 2.72 hrs only slightly reduces risk (1.16×). Daily screen time is the strongest predictor.

11. Summary of Key Insights

- Moderate and Severe addiction together dominate the labelled population.
- Weekend screen time and social media hours are strong behavioural markers of addiction severity.
- High notification volume (~134/day) and frequent app opens (~98/day) reinforce habitual checking.
- Daily screen time is the single strongest influencer of addiction status according to the key-influencers analysis.
- The 20–29 age group is the largest and highest-usage segment, making it the priority target for well-being interventions.

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- Sleep hours average 6.74 hrs – below recommended levels – highlighting a potential digital well-being risk area.
 - No significant gender disparity in average daily screen time, suggesting interventions can be largely gender-agnostic at the population level.

12. Conclusion

The Smartphone Addiction Analytics Dashboard successfully consolidates demographic, behavioural and lifestyle data into an interactive analytical experience. By delivering persona-specific views for product managers (Maya Chen), behaviour analysts (Daniel) and digital-health researchers (Dr. Priya Nair), the solution enables prioritisation of feature investment, early detection of high-risk segments, and evidence-based well-being recommendations.

All planned user stories across three sprints have been addressed. Data quality issues have been systematically identified and resolved. The resulting dashboard and reports provide actionable insights that support digital well-being programmes and product decision-making.